Q1. JEE Main 2026 (23 January Shift 1)

Consider the general reaction given below at 400 K

$x\,\mathrm{A}(\mathrm{g}) \rightleftharpoons y\,\mathrm{B}(\mathrm{g})$

The values of $\mathrm{K_p}$ and $\mathrm{K_c}$ are studied under the same condition of temperature but variation in x and y.

(i) $\mathrm{K_p} = 85.87$ and $\mathrm{K_c} = 2.586$ appropriate units

(ii) $\mathrm{K_p} = 0.862$ and $\mathrm{K_c} = 28.62$ appropriate units.

The values of x and y in (i) and (ii) respectively are:

(1) (i) $1, 3$ (ii) $2, 1$

(2) (i) $4, 1$ (ii) $4, 1$

(3) (i) $3, 1$ (ii) $3, 1$

(4) (i) $1, 2$ (ii) $2, 1$

Q2. JEE Main 2026 (23 January Shift 1)

For the following gas phase equilibrium reaction at constant temperature,

$\mathrm{NH_3}(\mathrm{g}) \rightleftharpoons 1/2\,\mathrm{N_2}(\mathrm{g}) + 3/2\mathrm{H_2}(\mathrm{g})$

if the total pressure is $\sqrt{3}$ atm and the pressure equilibrium constant $(\mathrm{K_p})$ is 9 atm, then the degree of dissociation is given as $\left(x \times 10^{-2}\right)^{-1/2}$. The value of x is ____. (nearest integer)

Q3. JEE Main 2026 (28 January Shift 2)

Observe the following equilibrium in a 1 L flask.

$\mathrm{A}(\mathrm{g}) \rightleftharpoons \mathrm{B}(\mathrm{g})$

At $\mathrm{T(K)}$, the equilibrium concentrations of A and B are 0.5 M and 0.375 M respectively. 0.1 moles of A is added into the flask and heated to $\mathrm{T(K)}$ to establish the equilibrium again. The new equilibrium concentrations (in M) of A and B are respectively

(1) $0.53, 0.4$.

(2) $0.742, 0.557$.

(3) $0.557, 0.418$.

(4) $0.367, 0.275$.

Q4. JEE Main 2026 (24 January Shift 2)

Consider the following gaseous equilibrium in a closed container of volume ' V ' at $\mathrm{T(K)}$.

$\mathrm{P_2}(\mathrm{g}) + \mathrm{Q_2}(\mathrm{g}) \rightleftharpoons 2\mathrm{PQ}(\mathrm{g})$

2 moles each of $\mathrm{P_2}(\mathrm{g}), \mathrm{Q_2}(\mathrm{g})$ and $\mathrm{PQ}(\mathrm{g})$ are present at equilibrium. Now one mole each of ' $\mathrm{P_2}$ ' and ' $\mathrm{Q_2}$ ' are added to the equilibrium keeping the temperature at $\mathrm{T(K)}$. The number of moles of P_2, Q_2 and PQ at the new equilibrium, respectively, are

(1) $1.66, 1.66, 1.66$

(2) $2.56, 1.62, 2.24$

(3) $2.67, 2.67, 2.67$

(4) $1.21, 2.24, 1.56$

Q5. JEE Main 2026 (22 January Shift 1)

Dissociation of a gas A_2 takes place according to the following chemical reaction. At equilibrium, the total pressure is 1 bar at 300 K.

$A_2(g) \rightleftharpoons 2A(g)$

The standard Gibbs energy of formation of the involved substances has been provided below:

Substance	$\Delta G_f^\circ / \text{kJ mol}^{-1}$
A_2	-100.00
A	-50.832

The degree of dissociation of $A_2(g)$ is given by $(x \times 10^{-2})^{1/2}$ where $x =$ ____. (Nearest integer).

[Given: $R = 8 \text{ J mol}^{-1} \text{ K}^{-1}, \log 2 = 0.3010, \log 3 = 0.48$]

Assume degree of dissociation is not negligible.

Q6. JEE Main 2026 (23 January Shift 2)

$X_2(g) + Y_2(g) \rightleftharpoons 2Z(g)$

$X_2(g)$ and $Y_2(g)$ are added to a 1 L flask and it is found that the system attains the above equilibrium at T(K) with the number of moles of $X_2(g)$, $Y_2(g)$ and Z(g) being 3,3 and 9 mol respectively (equilibrium moles). Under this condition of equilibrium, 10 mol of Z(g) is added to the flask and the temperature is maintained at T(K). Then the number of moles of Z(g) in the flask when the new equilibrium is established is ____. (Nearest integer)

Q7. JEE Main 2026 (21 January Shift 1)

For the reaction, $N_2O_4 \rightleftharpoons 2NO_2$, graph is plotted as shown below. Identify correct statements.

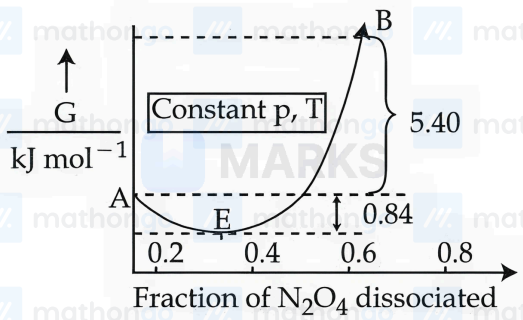
A. Standard free energy change for the reaction is $-5.40 \text{ kJ mol}^{-1}$.

B. As $\Delta G^\ominus$ in graph is positive, N_2O_4 will not dissociate into NO_2 at all.

C. Reverse reaction will go to completion.

D. When 1 mole of N_2O_4 changes into equilibrium mixture, value of $\Delta G^\ominus = -0.84 \text{ kJ mol}^{-1}$

E. When 2 mole of NO_2 changes into equilibrium mixture, $\Delta G^\ominus$ for equilibrium mixture is $-6.24 \text{ kJ mol}^{-1}$.



Choose the correct answer from the options given below :

(1) D and E only
(2) A and D only
(3) B and C only
(4) C and E only

ANSWERS AND SOLUTIONS

1. (4) **2.** 125 **3.** (3) **4.** (3) **5.** 33 **6.** 15 **7.** (1)

1. **(4)** The relationship between K_p and K_c is given by $K_p = K_c(RT)^{\Delta n_g}$, where $\Delta n_g = y - x$.
Given $T = 400$ K and $R = 0.0821$ L atm K^{-1} mol^{-1}.
The value of $RT = 0.0821 \times 400 = 32.84$.
For case (i): $K_p = 85.87$ and $K_c = 2.586$.
$\frac{K_p}{K_c} = \frac{85.87}{2.586} \approx 33.2$.
Since $33.2 \approx 32.84^1$, we have $\Delta n_g = y - x = 1$.
Checking options for (i):
(1) $y - x = 3 - 1 = 2$
(2) $y - x = 1 - 4 = -3$
(3) $y - x = 1 - 3 = -2$
(4) $y - x = 2 - 1 = 1$. This matches.
For case (ii): $K_p = 0.862$ and $K_c = 28.62$.
$\frac{K_p}{K_c} = \frac{0.862}{28.62} \approx 0.0301$.
Since $0.0301 \approx \frac{1}{32.84} = 32.84^{-1}$, we have $\Delta n_g = y - x = -1$.
Checking option (4) for (ii): $y - x = 1 - 2 = -1$. This also matches.
Thus, for (i) $x = 1, y = 2$ and for (ii) $x = 2, y = 1$.

2. **(125)** For the reaction: $NH_3(g) \rightleftharpoons \frac{1}{2}N_2(g) + \frac{3}{2}H_2(g)$

At $t = 0$: 1 mole, –, –
At equilibrium: $1 - \alpha$, $\frac{\alpha}{2}$, $\frac{3\alpha}{2}$
Total moles $= 1 + \alpha$

$$K_P = \frac{\left(\frac{\alpha}{2}\right)^{1/2}\left(\frac{3\alpha}{2}\right)^{3/2}}{(1-\alpha)} \cdot \left(\frac{P_T}{1+\alpha}\right)^1$$

Given $P_T = \sqrt{3}$ atm and $K_P = 9$ atm:

$$9 = \frac{\left(\frac{\alpha}{2}\right)^{1/2}\left(\frac{3\alpha}{2}\right)^{3/2}}{(1-\alpha)} \times \frac{(3)^{1/2}}{1+\alpha}$$

Simplifying: $9 = \frac{9\left(\frac{\alpha}{2}\right)^2}{1-\alpha^2}$

$1-\alpha^2=\frac{\alpha^2}{4}$

$\frac{5\alpha^2}{4}=1\Rightarrow\alpha^2=0.8$

$\alpha=(0.8)^{1/2}=\left(\frac{1}{0.8}\right)^{-1/2}=(125\times10^{-2})^{-1/2}$

So $x=125$.

3. **(3)** Initial equilibrium:
[A]=0.5 M, [B]=0.375 M; $K_c=\frac{0.375}{0.5}=0.75$
After adding 0.1 mol A: [A]=0.6 M, [B]=0.375 M; Q=0.625
Q < Kc, so equilibrium shifts right.
Let x be A that decomposes:
$\frac{0.375+x}{0.6-x}=0.75$
0.375+x=0.45-0.75x
1.75x=0.075; x≈0.0429
New [A]=0.6-0.0429=0.557 M; [B]=0.375+0.0429=0.418 M

4. **(3)** Initial equilibrium: 2 moles each of P_2, Q_2, PQ with $K=\frac{[PQ]^2}{[P_2][Q_2]}=\frac{4}{4}=1$.
After adding 1 mole each of P_2 and Q_2: initial composition is $(3,3,2)$. Reaction quotient $Q=\frac{4}{9}<K$, so reaction shifts right.
Let x moles of reactants combine:
$\frac{(2+2x)^2}{(3-x)^2}=1$
$2+2x=3-x$
$3x=1\rightarrow x=\frac{1}{3}$
Final moles: $P_2=3-\frac{1}{3}=2.67$, $Q_2=2.67$, $PQ=2+\frac{2}{3}=2.67$

5. **(33)** For the dissociation reaction $A_2(g)\rightleftharpoons 2A(g)$, first calculate the standard Gibbs free energy:
$\Delta G^\circ{}_{rxn}=2(-50.832)-(-100.00)=-1.664$ kJ/mol.
Using $\Delta G^\circ=-RT\ln K_p$:
$-1664=-8\times300\times\ln K_p$, giving $K_p=e^{0.6933}=2$.
For dissociation with degree of dissociation α, starting with 1 mole of A_2 at 1 bar total pressure:
At equilibrium, moles are $(1-\alpha)$ for A_2 and 2α for A, with total $(1+\alpha)$ moles.
$K_p=\frac{4\alpha^2}{1-\alpha^2}=2$.
Solving: $2(1-\alpha^2)=4\alpha^2$ gives $\alpha^2=1/3$ or $\alpha=0.577$.
Since degree of dissociation is $(x\times10^{-2})^{1/2}$:
$(x\times10^{-2})^{1/2}=0.577$ gives $x\times10^{-2}=0.333$, so $x=33.3\approx33$.

6. **(15)** The equilibrium constant for the reaction $X_2(g) + Y_2(g) \rightleftharpoons 2Z(g)$ is calculated from the initial equilibrium state:

$K_c = \frac{[Z]^2}{[X_2][Y_2]} = \frac{9^2}{3\times3} = 9$ mol/L.

When 10 mol of Z(g) is added, the new concentrations are: $[X_2] = 3$, $[Y_2] = 3$, $[Z] = 19$ mol.

The reaction quotient is $Q = \frac{19^2}{3\times3} = 40.11$.

Since $Q > K_c$, the equilibrium shifts left (toward reactants).

Let x mol of Z decompose. At the new equilibrium: $[X_2] = 3 + \frac{x}{2}$, $[Y_2] = 3 + \frac{x}{2}$, $[Z] = 19 - x$.

Applying the equilibrium expression: $\frac{(19-x)^2}{(3+\frac{x}{2})^2} = 9$.

Taking the square root: $\frac{19-x}{3+\frac{x}{2}} = 3$.

Solving: $19 - x = 9 + \frac{3x}{2}$, which gives $10 = \frac{5x}{2}$, so $x = 4$.

The moles of Z at new equilibrium = $19 - 4 = 15$ mol.

7. **(1)** From the graph of G vs fraction of N_2O_4 dissociated:

At $x = 0$ (pure N_2O_4): $G = 0$

At equilibrium ($x \approx 0.4$): $G = -0.84$ kJ/mol (minimum)

At $x = 1$ (complete dissociation): $G = 5.40$ kJ/mol

Statement A: $\Delta G^\circ = 5.40$ kJ/mol (positive), not -5.40 kJ/mol. False.

Statement B: Even with positive ΔG°, equilibrium is established with partial dissociation. False.

Statement C: Reverse reaction reaches equilibrium, not completion. False.

Statement D: $\Delta G = G_{eq} - G_{N_2O_4} = -0.84 - 0 = -0.84$ kJ/mol. True.

Statement E: $\Delta G = G_{eq} - G_{NO_2} = -0.84 - 5.40 = -6.24$ kJ/mol. True.